

3.

(a) BRIEF SUMMARY:

The paper introduces a set of regression models built specifically for data where the dependent variable is ordinal, i.e. they have a logical order.

Instead of just assigning numerical scores to these categories which assumes the distance between each category is the same, it assumes that there is a latent variable that we can't measure directly. The model treats the target variable (discrete and ordered) as contiguous blocks / intervals along this hidden continuous scale. These blocks are separated by unknown boundary markers called "cut points" (θ_j). By this way, we analyze the data based purely on its order without forcing fake scores on it.

The paper specifically highlights two models that are useful for this: proportional odds model & proportional hazards model. Essentially these models help us measure a "stochastic ordering" i.e. if a specific factor shifts the response higher or lower, the model captures the shift consistently across all category boundaries.

The paper also applies the model to real world examples: (i) it shows how you can quantify exactly how much being a bacteria carrier increases the odds of having enlarged tonsils across multiple size categories; (ii) it also shows how the

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severity of disturbed dreams smoothly decreases by a specific factor as boys age from 5 to 15. The model essentially summarizes this overall relationship to a single parameter that captures how the distribution shifts across groups making the results much more useful for quantitative descriptions.

LIKELIHOOD & ODDS RATIO:

Ordinal regression relies on cumulative probabilities rather than probability of falling into a certain class.

If $\pi_j(x) :=$ probability of being in category j , the cumulative probability is defined as:

$$\delta_j(x) = \pi_1(x) + \pi_2(x) + \dots + \pi_j(x)$$

The odds of being in category j or lower is written as $K_j(x)$.

$$K_j(x) = \frac{\delta_j(x)}{1 - \delta_j(x)}$$

In proportional odds model, this is expressed using linear logistic equation

$$\log[K_j(x)] = \theta_j - \beta^T x$$

$\therefore$ the odds ratio for two different data points x_1 & x_2 is

$$\frac{K_j(x_1)}{K_j(x_2)} = \exp\{\beta^T (x_2 - x_1)\}$$

which is completely independent of the specific category j being looked at

To calculate likelihood, we start with standard multinomial likelihood, $\pi_1^{n_1} \dots \pi_k^{n_k}$ and transform it into cumulative totals.

If $n_j :=$ count in category j

the cumulative sums are defined as:

$$R_1 = n_1$$

$$R_2 = n_1 + n_2 \text{ and so on}$$

$$R_k = n$$

The likelihood is factored into product of $k-1$ conditional probabilities. For instance, the first part is the probability that the first two cells split in ratio $R_1 : R_2 - R_1$,

$$\text{i.e. } \left\{ \left(\frac{r_1}{r_2} \right)^{R_1} \left(\frac{r_2 - r_1}{r_2} \right)^{R_2 - R_1} \right\} \Rightarrow \text{likelihood naturally takes in the cumulative ordered nature of the data.}$$

Multiclass classification handles likelihood & odds ratio differently $\because$ the categories do not have a strict order.

$$\therefore p(y=k|x) = \frac{\exp(w_k^T x)}{\sum_{i=1}^K \exp(w_i^T x)} \quad [\text{using the softmax function}]$$

For odds ratio,

$$\frac{p(y=k|x)}{p(y=j|x)} = \frac{\exp(w_k^T x)}{\exp(w_j^T x)} = \exp[(w_k - w_j)^T x]$$

Unlike this, in ordinal regression, because of the proportional odds assumption, it uses a single shared set of weights, β , to describe the shift.

DIFFERENCE FROM REGRESSION PROBLEMS

In standard regression problems, we want to predict a continuous, real-valued output $y \in \mathbb{R}$.

For linear regression

$$y = w^T x + \epsilon \quad \text{where } \epsilon \sim \mathcal{N}(0, \sigma^2)$$

polynomial regression,

$$y = \sum_{i=0}^M w_i x^i \quad \text{where } M := \text{order of the polynomial}$$

In both cases, we try to predict the exact numerical value.

Ordinal regression works differently. The target variable falls into set of discrete categories rather than being continuous. Because a std regression line can't be fit into qualitative bins, ordinal models apply a link function (like logit or probit). $\therefore$ instead of estimating continuous outputs directly, the ordinal model ~~settles~~ estimates unknown boundary cut points (θ_j) that divide the ordered categories on the ~~at~~ hidden continuous scale.